Hall Ticket Number:

MR-21

Code: B415CS4S

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. IV Semester Supplementary Examinations, July 2025

Java Programming

(Common to CSE and IT)

Time: 3 Hours | ***Answer ALL Questions*** | Max. Marks: 70

Part-A (10 x 2 = 20 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	Define Dynamic method dispatch.	2	1	1	1,2
1. b)	List the different types of inheritance.	2	1	1	1,2
1. c)	Multiple inheritance can be achieved using interfaces. How?	2	1	1	1,2
1. d)	Define throw, throws, finally.	2	1	2	1,2
1. e)	Define Serialization and Enumerations.	2	1	3	1,2
1. f)	Why threads need to be synchronized in java.	2	1	4	1,2
1. g)	Compare Hashtable and Dictionary.	2	2	3	1,2
1. h)	List the operations on Stack.	2	1	3	1,2
1. i)	Draw the Applet life Cycle.	2	2	5	1,2
1. j)	List out the limitations of AWT.	2	1	5	1,2

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Define Class. Explain declaration of object with an example.	6	2	1	1,2,3
2. b)	Show the syntax to declare an object for bank class defined with two relevant attributes and two methods.	4	1	1	1,2,3
	OR				
2. c)	Illustrate the usage of super keyword with an example.	4	2	1	1,2,3
2. d)	Specify the purpose of a constructor. Explain different types of constructors with examples.	6	3	1	1,2
3. a)	Explain nested interfaces with a suitable example.	4	2	2	1,2
3. b)	Define CLASSPATH. Analyze the different access protection mechanisms used in packages?	6	4	2	1,2
	OR				
3. c)	List any three built-in exceptions in java with examples.	4	1	2	1,2
3. d)	Write a java program to define user-defined exceptions.	6	3	2	1,2,3
4. a)	Explain different console input and output methods with examples.	4	2	3	1,2
4. b)	Write a java program to read a string and sort the string using streams.	6	3	3	1,2,3
	OR				
4. c)	Compare thread-based and process-based multitasking.	4	3	4	1,2
4. d)	Write a java program to copy one file to another file.	6	3	3	1,2,3
5. a)	Explain about String Tokenizer with an example.	5	2	3	1,2,3
5. b)	Explain LinkedList with an example.	5	2	3	1,2,3
	OR				
5. c)	Explain Collection classes with an example.	4	2	3	1,2
5. d)	Write a Java program to read your name and access the content using iterator.	6	3	3	1,2,3
6. a)	Write a simple Applet program to display welcome message.	6	3	5	1,2,3
6. b)	Explain about MVC architecture in detail.	4	2	5	1,2
	OR				
6. c)	Write a java program to handle mouse and keyboard events.	6	3	5	1,2,3
6. d)	Explain different layout managers.	4	2	5	1,2

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome

Hall Ticket Number:

MR-21

Code: B415CB4S

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. IV Semester Supplementary Examinations, July 2025

Object Oriented Programming using Java

(Common to CSB, CSM and CSD)

Time: 3 Hours | ***Answer ALL Questions*** | Max. Marks: 70

Part-A (10 x 2 = 20 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	What is JVM in Java?	2	1	1	1
1. b)	What is the difference between overloading and overriding?	2	1	1	1
1. c)	Write the differences between throw and throws clauses.	2	1	2	2
1. d)	What is exception handling? Explain an example of exception handling in the case of division by zero.	2	2	2	2
1. e)	What is the difference between a thread and a process in Java?	2	1	3	2
1. f)	Write down the advantages of Java Serialization.	2	2	3	1
1. g)	What is a Collections Framework? Benefits of the Java Collections Framework.	2	1	4	1
1. h)	Write down six useful methods of the StringTokenizer class.	2	2	4	1
1. i)	Draw the life cycle of an applet.	2	2	5	1
1. j)	Why swing components are preferred over AWT components?	2	2	5	2

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Explain the basic concepts of object-oriented programming.	5	1	1	1
2. b)	Write Java Program to find Transpose of Matrix.	5	3	1	2
	OR				
2. c)	What is string in Java? What are String handling functions in Java. Write a Java Program to Merge Two String Arrays.	5	3	1	1
2. d)	Write the differences between interface and abstract class by giving examples.	5	2	1	2
3. a)	Can inheritance be applied between interfaces? Justify your answer.	5	4	2	1
3. b)	With a suitable Java program explain user-defined exception handling.	5	2	2	2
	OR				
3. c)	Design an interface called Shape with methods draw () and getArea(). Further design two classes called Circle and Rectangle that implements Shape to compute area of respective shapes. Use appropriate getter and setter methods. Write a java program for the same.	5	3	2	2
3. d)	What is meant by re-throwing exception? Discuss a suitable scenario for this.	5	2	2	2
4. a)	How to achieve synchronization among threads? Write suitable code.	5	2	3	2
4. b)	Explain the file management using File class.	5	2	3	1
	OR				
4. c)	Write a program to create three threads in your program and Context switch among the threads using sleep functions.	5	3	3	3
4. d)	Explain about Generic Classes & Method with an example.	5	2	3	1

P.T.O.

5. a)	Explain methods associated with Set Interfaces of Collection framework with Examples.	5	3	4	1
5. b)	With syntax, explain the following utility classes. i) StringTokenizer ii) Date and Calendar iii) Scanner.	5	1	4	1
	OR				
5. c)	Differentiate between ArrayList and Vector. List the methods of Stack class.	5	4	4	2
5. d)	With syntax, explain the following utility classes. i) HashSet ii) TreeSet iii) Hash Table.	5	1	4	1
6. a)	Write an applet program to handle all mouse events	5	3	5	2
6. b)	Explain briefly components and containers used in swings. Write a program to create three buttons with caption OK, SUBMIT, CANCEL.	5	3	5	3
	OR				
6. c)	Create a swing applet that has two buttons named as "CSBS" and "IT" and When either of the button is pressed, it should display "CSBS is pressed" and "IT is pressed respectively".	5	3	5	3
6. d)	What are the various layout managers used in Java? Give java code for at least two types.	5	2	5	1

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome